

Class 17: RNASeq Pop Scale Analysis

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Population Scale Analysis

Q13: Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes.

```
expr <- read.table("rs8067378_ENSG00000172057.6.txt")
head(expr)
```

	sample	geno	exp
1	HG00367	A/G	28.96038
2	NA20768	A/G	20.24449
3	HG00361	A/A	31.32628
4	HG00135	A/A	34.11169
5	NA18870	G/G	18.25141
6	NA11993	A/A	32.89721

```
table(expr$geno)
```

A/A	A/G	G/G
108	233	121

```
median(expr$exp[expr$geno == "A/A"])
```

```
[1] 31.24847
```

```
median(expr$exp[expr$geno == "A/G"])
```

```
[1] 25.06486
```

```
median(expr$exp[expr$geno == "G/G"])
```

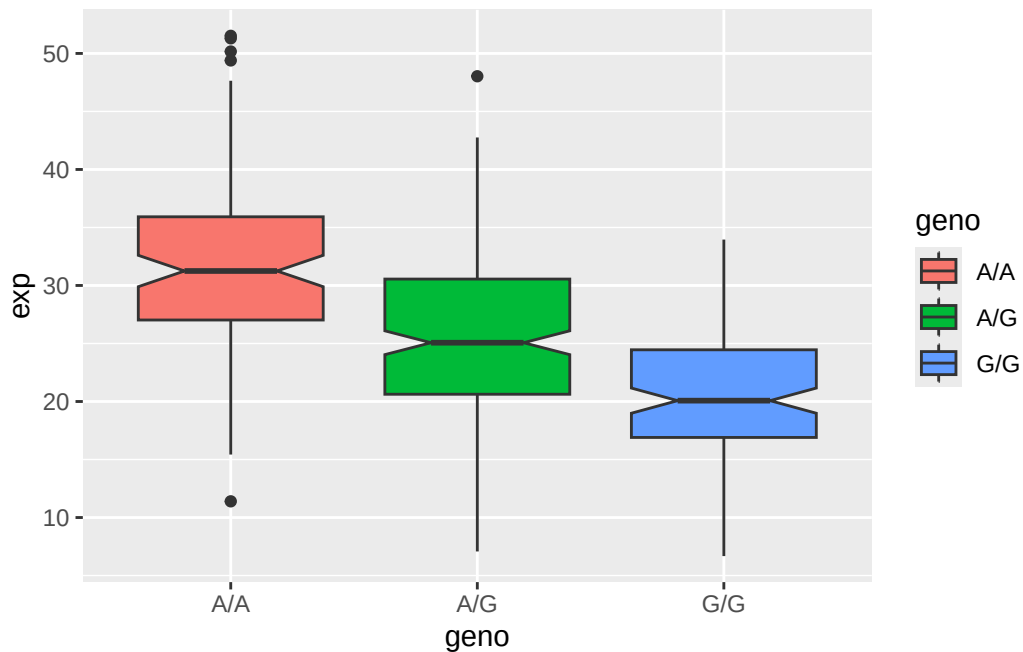
```
[1] 20.07363
```

The sample size for the A/A is 108, A/G is 233, and G/G is 121. The median expression levels for A/A is 31.24847, A/G is 25.06486, and G/G is 20.07363.

Q14: Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3?

```
library(ggplot2)
```

```
ggplot(expr) +aes(geno, exp, fill = geno) +geom_boxplot(notch = T)
```



Based on the histogram, this specific SNP does have an effect on the expression of ORMDL3. In the presence of homozygous allele SNPs there is lower gene expression of ORMDL3 than the wildtype (A/A) and heterozygous alleles. The heterozygous allele (A/G) also result in a lowered expression of ORMDL3, but not as low as the homozygous SNP (G/G)